Fiscal Year:	FY 2019	**Task Last Updated:**	FY 01/15/2019
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Project Title:	Discovery of Human Radiation-protection Genes and Pathways		
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Program/Discipline--Element/Subdiscipline:	TRISH--TRISH		
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Comments:			
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No. of PhD Candidates:	1	**No. of Master' Degrees:**	
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No. of Bachelor's Candidates:	1	**Monitoring Center:**	TRISH
Contact Monitor:		**Contact Phone:**	
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Key Personnel Changes/Previous PI:			
COI Name (Institution):	Miller, Kyle Ph.D. (University of Texas, Austin)		
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Summary: Discovery of human radiation-protection genes and pathways
Rationale: Individuals in high-radiation environments, including space, suffer DNA damage, which increases their susceptibility to cancer, among other diseases. The levels of DNA damage accumulated in cells can be used to measure the extent to which cells succumb to, or alternatively resist, the deleterious consequences of radiation. This project leverages the only known collection of genes that confer lower-than-normal levels of spontaneous DNA damage to cells to identify proteins that confer resistance to exogenous proton-beam (space-relevant) radiation.

Unique resource: We discovered 231 Escherichia coli (bacterial) genes that alter levels of DNA damage in cells when overproduced—208 that increase and 23 that decrease DNA damage, the latter of interest to radiation resistance. The

Task Description:	human-gene relatives of the bacterial “damage-up” genes also increased DNA damage when overproduced in human cells, and are highly significantly overrepresented among known cancer-driving genes. These data demonstrate the relevance and power of conserved bacterial genes for discovery of important human biology of DNA damage. Plan: We will explore the 23 E. coli DNA “damage-down” genes and their human homologs and analogs for their ability, when overproduced, to protect cells from exogenously applied proton-induced DNA damage. We will identify—(1) which E. coli genes confer resistance to proton-induced DNA damage; (2) what kinds of DNA damage they reduce; (3) which of their human-gene relatives confer resistance to proton-mediated DNA damage when overproduced, and guided by the bacterial results, test hypotheses for how they do so. Deliverables: Identities of bacterial and human proteins that protect cells from DNA damage induced by proton beams, a proxy for protection from radiation generally, and some of the mechanisms by which they do so. The human proteins and pathways of radiation resistance, when understood, can be considered as potential targets for, or models for design of, drugs for protection from radiation.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	We expect the results of this to map a set of proteins the pharmacological manipulation of which could potentially promote enhanced radiation resistance to astronauts in space, and may also prevent aging and age-related diseases such as cancer.
Task Progress:	Overview: This project leverages our discovery of a set of conserved bacterial proteins that, when overproduced, confer reduced cellular levels of endogenous DNA damage. We use these for prediction and potential discovery of similar human proteins that may promote resistance to exogenous DNA-damaging agents including space-relevant radiation. The project goals are to discover which of 23 Escherichia coli (bacterial) endogenous DNA Damage-Suppressing Proteins (DDSPs) can reduce DNA damage from exogenous sources, some of the mechanisms by which they do so, and which of their human counterparts may do this. The ultimate goal is to understand whether pharmacologically increased production of some/any of the human candidate proteins might protect astronauts from radiation in space, and also potentially extend healthy life-span and prevent cancer in people on Earth. We have made significant progress on the two aims: Aim 1: Discovery of E. coli DNA damage-down proteins that protect bacteria from exogenous DNA damage when overproduced. Aim 2: Discovery of human homologs, analogs, and pathways that protect human cells from exogenous DNA damage when overproduced. Key findings: 1) Dose-response curves were generated and used to determine appropriate experimental conditions for treatment of E. coli cells with the exogenous DNA-damaging agent -- the radiometric drug, phleomycin. For exogenous DNA damage with gamma rays, dose response curves are in progress and close to completion. 2) We have discovered five E. coli proteins that protect cells from DNA damage caused by an exogenous DNA-damaging agent when overproduced. 3) We have cloned full-length green fluorescent protein (GFP) fusions of 82 human candidate DDSP genes, and determined the subcellular localization patterns of 72 of them -- a prerequisite to testing these proteins in human cells. 4) Dose-response curves were generated and used to determine appropriate experimental conditions for treatment of one of two human cell lines with the exogenous DNA-damaging agent -- X-rays. The dose-responsive curves with X-rays for a separate human cell line are in progress. 5) Preliminary (tentative) data from an initial screen of 67 human homologs (proteins of similar amino-acid sequence) and analogs of the bacterial DDSPs (proteins that function similarly to E. coli proteins but do not share amino-acid sequence identity) indicate that most may show reduced endogenous DNA-damage levels. 6) From a separate project, serendipitously, we discovered and have fully validated one human protein that reduces spontaneous endogenous DNA damage in human cells when overproduced -- is a genuine human DDSP (hDDSP). 7) In additional preliminary data, we have found that overproduction of the sole validated hDDSP can protect human cells from exogenous radiation-induced DNA damage. Impact: 1) Determination of conditions for bacterial experiments is complete for one DNA-damaging agent and nearly complete for another. This lets the project proceed. 2) Our initial hypothesis that DDSPs might protect cells from exogenous as well as endogenous DNA damage has been supported in E. coli. This supports the rationale for the project. 3) A major swat of laborious construction work required for testing the human proteins has been achieved. 4) Determination of conditions for experiments using X-rays in one human cell line is complete, and has been initiated for another human cell line. This lets the project proceed. 5) Preliminary data, which still require repetition and validation, suggest that many human candidate proteins predicted by the E. coli network may reduce endogenous DNA damage levels (be hDDSPs) when overproduced, potentially paving the way to discovery of radiation-protection proteins among them. Much work remains to validate these, but the first results are promising. 6) We discovered our first validated genuine hDDSP, which reduces levels of endogenous DNA damage in human cells,

	a promising candidate for protection from exogenous DNA-damaging agents. 7) Preliminary data, which must be replicated and validated, suggest that overproduction of the first demonstrated hDDSP may protect human cells from DNA damage caused by exogenous radiation. If validated, this would be a candidate for a health-promoting and radiation-protection protein potentially useful for human wellness and protection of astronauts. Results: The key results and their significance are outlined in the two sections above. Several more human proteins are being prepared for testing -- their genes are being cloned, several experimental assay systems in bacteria and human cells are being developed for achieving the goals of this project, results are being repeated, validated in different assays, and many more bacterial and human proteins are being tested in several assays for achieving the goals of this work.
Bibliography Type:	Description: (Last Updated: 07/30/2025)
Awards	Chiu L. "Dissertation Writing Fellowship, The University of Texas at Austin, January, 2018." Jan-2018
Awards	Rosenberg SM. "Public Lecture, Arizona State University (honorary lecture), April 2018." Apr-2018
Awards	Rosenberg SM. "Dean's Distinguished Seminar, University of Colorado School of Medicine, Aurora, CO (honorary lecture), September 2018." Sep-2018
Awards	Rosenberg SM. "Keynote lecture, Ohio State University Biomedical Sciences Graduate Program Retreat, Columbus, OH, December 2017." Dec-2017
Awards	Xia J. "Outstanding Reviewer. DNA Repair, Elsevier, November 2017." Nov-2017